FITBUDDY

AI Fitness Plan Generator Using Gemini Models

TEAM LEADER

Sreesh .V

NM ID

761C7470CE1315F9C70461CBA349136

Project Report

An AI-powered web application for generating general, personalized fitness plans using Google Gemini models.

1. Project Title

FitBuddy – AI Fitness Plan Generator Using Gemini Models

2. Abstract

FitBuddy is an Artificial Intelligence-based fitness planning application that uses Google Gemini models to generate personalized fitness and workout plans. The main purpose of FitBuddy is to help users create structured workout routines according to their fitness goals, experience level, available time, preferred workout type, and available equipment.

The application can generate weekly workout schedules, exercise suggestions, repetitions, workout duration, rest periods, and general exercise explanations. Users can also interact with an AI assistant to ask questions about their generated plan.

FitBuddy is intended as a general fitness-planning and educational tool. AI-generated recommendations should not be treated as medical advice, and users with health conditions or injuries should consult an appropriate professional.

3. Introduction

Fitness and physical activity play an important role in maintaining a healthy lifestyle. However, many people find it difficult to create a proper workout routine because they may not know which exercises to perform, how frequently to exercise, or how to organize their weekly schedule.

Different people have different fitness requirements. A beginner may require a simple workout routine, while an experienced person may prefer a more challenging routine. Some users may have access to gym equipment, while others may only be able to exercise at home.

FitBuddy addresses this problem by using Artificial Intelligence to generate customized fitness plans. The user provides basic information through a web interface, and the Gemini model processes the information to generate a suitable general-purpose fitness plan.

4. Problem Statement

Many people want to follow a regular fitness routine but face problems such as:

- Lack of knowledge about suitable exercises.
- Difficulty creating a weekly workout schedule.
- Limited access to personal trainers.
- Difficulty adapting workouts to available equipment.
- Difficulty maintaining a consistent routine.
- Large amounts of fitness information available online.
- Difficulty selecting information suitable for their experience level.

Therefore, there is a need for an easy-to-use application that can generate structured fitness plans based on individual user preferences.

5. Proposed Solution

FitBuddy provides an AI-powered solution where users enter their fitness requirements and receive a generated workout plan.

- Collects user information.
- Validates the information.
- Sends the information to the backend.
- Creates a structured prompt.
- Sends the prompt to the Gemini model.
- Receives the generated response.
- Formats the response.
- Displays the fitness plan to the user.

6. Objectives

Primary Objectives

- Develop an AI-based fitness planning application.
- Integrate Google Gemini models into a practical application.
- Generate personalized workout plans.
- Create an interactive user interface.
- Create weekly workout schedules.
- Provide general exercise information.

Secondary Objectives

- Demonstrate Generative AI integration.
- Develop a FastAPI backend.
- Implement frontend-backend communication.
- Provide error handling and validation.
- Create a scalable project structure.
- Provide a foundation for future mobile and wearable integrations.

7. Scope of the Project

FitBuddy can be used to generate general fitness plans based on user preferences.

- Beginner fitness plans.
- Intermediate workout plans.
- General fitness routines.
- Home workouts.
- Gym-based routines.
- Bodyweight workouts.
- Flexible workout schedules.
- Exercise explanations.

- General fitness questions.
- Weekly workout organization.

Future Scope

- Progress tracking.
- User accounts.
- Workout history.
- Mobile application.
- Wearable integration.
- Exercise videos.
- Calendar reminders.
- Multilingual AI assistance.

8. Key Features

8.1 User Profile

- Name.
- Age.
- Fitness experience.
- Fitness goal.
- Workout preference.
- Available equipment.
- Workout duration.
- Number of workout days.

8.2 AI Fitness Plan Generator

The Gemini model generates a fitness plan based on the information provided by the user. The generated plan can contain exercise names, sets, repetitions or duration, rest periods, workout days, and general instructions.

8.3 Weekly Workout Schedule

FitBuddy can organize exercises into a weekly structure. For example: Monday – Full Body; Tuesday – Light Activity/Rest; Wednesday – Lower Body; Thursday – Rest; Friday – Upper Body; Saturday – Core/Mobility; Sunday – Rest. The exact plan is generated according to the user's inputs.

8.4 Exercise Information

- Exercise name.
- General purpose.
- Basic instructions.
- Suggested duration or repetitions.
- Difficulty level.
- Rest period.

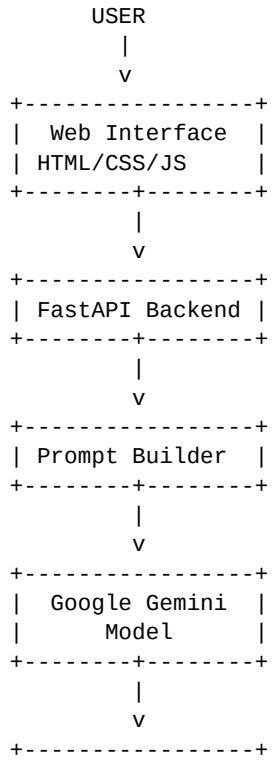
8.5 AI Chat Assistant

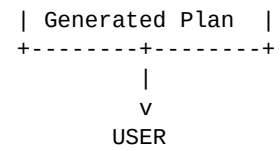
- Explain an exercise.
- Explain how to organize a workout week.
- Identify general equipment requirements.
- Explain the generated plan.

9. Technology Stack

Component	Technology
Frontend	HTML
Styling	CSS
Client-side Logic	JavaScript
Backend	Python
API Framework	FastAPI
AI	Google Gemini
Database	SQLite / MySQL
API	REST API
IDE	Visual Studio Code
Version Control	Git / GitHub
Browser	Chrome / Edge

10. System Architecture





11. System Workflow

Step 1 – Open Application

The user opens the FitBuddy website.

Step 2 – Enter Information

The user enters their fitness preferences.

Step 3 – Submit Form

The information is submitted to the backend.

Step 4 – Backend Processing

FastAPI receives and validates the data.

Step 5 – Prompt Creation

The backend creates a structured prompt using the user's information.

Step 6 – Gemini Processing

The Gemini model processes the prompt.

Step 7 – Generate Plan

Gemini generates the requested fitness plan.

Step 8 – Response Processing

The backend receives and formats the AI response.

Step 9 – Display

The final plan is displayed on the FitBuddy interface.

12. Epic 1 – Model Selection and Architecture

A. Select AI Models

The main AI component of FitBuddy is the Google Gemini model. Gemini is responsible for natural-language processing and generation.

- Understanding user requirements.
- Generating workout plans.
- Explaining exercises.
- Answering general fitness questions.
- Modifying plans according to preferences.
- Generating structured text.

Example Input

```
Goal: General Fitness
Experience: Beginner
Workout Days: 4
Duration: 30 minutes
Equipment: No equipment
```

Example AI Output

```
Day 1:
Full Body Workout

Exercise 1: Bodyweight Squat
Exercise 2: Wall Push-up
Exercise 3: Glute Bridge
Exercise 4: Plank

Include appropriate warm-up, rest periods,
and cool-down according to the user's needs.
```

13. Epic 2 – Core Functionalities Development

A. Module Implementation

Module 1 – User Input

Collects information from the user.

Module 2 – Input Validation

Checks whether required information has been entered correctly.

Module 3 – Prompt Generator

Converts user information into a structured Gemini prompt.

Module 4 – AI Generator

Communicates with the Gemini API.

Module 5 – Plan Formatter

Converts the AI response into a readable format.

Module 6 – Workout Dashboard

Displays the generated workout plan.

Module 7 – AI Assistant

Allows the user to interact with Gemini.

14. Backend API with FastAPI

FastAPI provides the backend services for FitBuddy.

```
POST /generate-plan
Generates a fitness plan.

POST /chat
Processes general AI questions.

GET /health
```

```
Checks whether the backend is working.
```

15. API Request Flow

```
User
 ↓
Frontend
 ↓
POST /generate-plan
 ↓
FastAPI
 ↓
Validate Data
 ↓
Create Prompt
 ↓
Gemini API
 ↓
AI Response
 ↓
FastAPI
 ↓
Frontend
 ↓
Display Plan
```

16. Epic 3 – Frontend Development

A. Build Web Interface

- Home Page – FitBuddy logo, project title, description, and start button.
- Fitness Form – Age, goal, experience, workout days, duration, equipment, and location.
- Result Page – Weekly schedule, daily workouts, exercise information, duration, and rest periods.
- AI Assistant Page – Interactive chat interface.

Frontend Structure

```
FitBuddy/
■
■■■ index.html
■■■ style.css
■■■ script.js
■
■■■ backend/
■   ■■■ main.py
■   ■■■ models.py
■   ■■■ services.py
■
■■■ database/
■   ■■■ database.db
■
■■■ README.md
```

17. Database Design

User Table

Field	Description
user_id	Unique user ID

Field	Description
name	User name
age	User age
goal	Fitness goal
experience	Experience level
workout_days	Preferred days
duration	Workout duration
equipment	Available equipment

Workout Table

Field	Description
workout_id	Unique workout ID
user_id	Associated user
date	Creation date
plan	Generated plan
duration	Workout duration

18. Epic 4 – Testing and Deployment

A. Run Locally

- Install Python.
- Install Visual Studio Code.
- Create the project folder.
- Create and activate a virtual environment.
- Install required Python packages.
- Configure the Gemini API key securely.
- Start the FastAPI server.
- Open the application in a web browser.

```
mkdir FitBuddy
cd FitBuddy
python -m venv venv
uvicorn main:app --reload
```

19. Functional Testing

Test Case	Input / Condition	Expected Result
1 – Valid User Input	Complete fitness information	AI generates a structured fitness plan.
2 – Missing Information	Required fields empty	Validation message is displayed.
3 – Gemini API Error	API unavailable	Appropriate error message is displayed.
4 – Experience Level	Beginner / Intermediate	Response reflects selected experience level.
5 – Equipment	Different equipment choices	Plan considers stated equipment.

20. Non-Functional Requirements

- Performance – Process requests efficiently.
- Usability – Keep the interface simple and easy to understand.
- Reliability – Handle API errors without crashing.
- Security – Do not expose API credentials in frontend code.
- Scalability – Allow future features to be added.
- Maintainability – Organize code into separate modules.

21. Security

- Store API keys in environment variables.
- Do not place API keys in HTML or JavaScript.
- Validate user inputs.
- Protect backend endpoints.
- Avoid storing unnecessary personal information.
- Use HTTPS when deployed.
- Apply appropriate authentication if user accounts are introduced.

22. Responsible AI

Because FitBuddy generates fitness-related content using AI, responsible AI practices are important.

- Clearly identify AI-generated information.
- Avoid presenting the AI as a medical professional.
- Avoid diagnosing medical conditions.
- Avoid guaranteeing specific health outcomes.
- Encourage professional guidance where appropriate.
- Handle potentially unsafe requests carefully.
- Provide general rather than medically individualized advice.

23. Advantages

- Personalized – Plans can be generated according to user preferences.
- Interactive – Users can communicate with the AI assistant.
- Fast – A workout plan can be generated quickly.
- Accessible – Users can access the application through a web browser.
- Flexible – Plans can be generated for different schedules and equipment levels.
- Expandable – The system can later be developed into a mobile application.

24. Limitations

- AI-generated information may contain errors.

- Internet access may be required.
- Gemini API availability and usage limits may apply.
- AI cannot replace professional fitness or medical assessment.
- Quality of generated plans depends partly on information provided by the user.
- The application may require additional validation before production use.

25. Future Enhancements

- Mobile application.
- User authentication.
- Progress tracking.
- Workout history.
- Exercise videos.
- AI voice assistant.
- Wearable integration.
- Workout reminders.
- Multilingual support.
- Advanced dashboard with progress charts and workout statistics.

26. Expected Output

```
User Information
       ↓
AI Processing
       ↓
Personalized Fitness Plan
       ↓
Weekly Workout Schedule
       ↓
Exercise Information
       ↓
AI Assistance
```

The final application should provide users with an interactive platform for generating and viewing general fitness plans.

27. Project Outcome

The successful implementation of FitBuddy will demonstrate Artificial Intelligence integration, Google Gemini API integration, Python programming, FastAPI backend development, frontend development, REST API communication, database integration, prompt engineering, AI-generated content processing, and responsible AI practices.

28. Conclusion

FitBuddy – AI Fitness Plan Generator Using Gemini Models is an AI-based application that demonstrates how Generative AI can be used to create personalized fitness-planning experiences.

The application combines a user-friendly frontend, FastAPI backend, and Google Gemini models to generate structured workout plans based on user requirements. The system provides an interactive

approach to fitness planning and can be expanded with progress tracking, mobile applications, wearable integration, voice interaction, and multilingual support.

Overall, FitBuddy provides a strong example of applying modern AI technology to a practical real-world application while maintaining appropriate limitations and responsible-use considerations.

29. Team Details

Team Leader: Sreesh .V
NM ID: 761C7470CE1315F9C70461CBA349136

Project: FitBuddy – AI Fitness Plan Generator Using Gemini Models

30. References

• Google Gemini API documentation.
• FastAPI documentation.
• Python documentation.
• HTML, CSS and JavaScript documentation.
• SQLite / MySQL documentation.
• General exercise and fitness education resources from reliable professional organizations.

Note: The exact Gemini model name and API/SDK syntax should be selected according to the current Google AI documentation during implementation, since model availability and API interfaces can change.